Software Engineering

04GSPOV

Course Information

What

- What will I learn in this course?

The software ‘factory’

software

Why

I can write a {C, Java, Python, Javascript, PHP, ….} program, what else should I need to know?

Software engineering is about software as a **commercial, industrial product**

What was the largest program you coded?

```
# taking user input

ch =  ("Enter a character: ")
If ((ch>='a' and ch<= 'z') or (ch>='A' and ch<='Z')):
            (ch, "is an Alphabet")
else:
            (ch, "is not an Alphabet")
```

This program, 5 LOC = Lines Of Code

SoftEng http://softeng.polito.it

- www.menti.com
- Code **7984 3526**

Size of ‘real’ programs

- Linux OS: > 15 M LOC
- Android OS: > 12 M LOC
- Windows : > 50 M LOC
- Boeing 787: > 7 M LOC
- Car (high level models): > 100M LOC
- ERP : > 1500 RDB tables

1 M = 1 MILLION

Why

Software as an industrial product

- Is large, and requires a **team**, not a single programmer

  need to share programs and documents

  → configuration management, repository tools

  need to manage the team

  → project management, team management

Why

Software as an industrial product

- Is made for a **customer**, who has **no notion of computing**

    need to understand what the customer wants / needs

        → requirement engineering

    need to define and respect deadlines and contracts

        → project management

    need to assure a defined level of quality

        → testing

Why

Software as an industrial product

- Is usually a **large and complex** product

  need to master the complexity, and implement what was promised to the customer

  → software design

  → project management

Why

Software as an industrial product

- Is an asset with a **great financial value** that must last years

  need to maintain and evolve the product for years

  → software maintenance

  → software design

  → testing

  → configuration management

Why

Software as an industrial product

- Is an asset with **critical effect on the customer's business**

    need to avoid critical defects

    → software testing

    → software quality

Critical failures

- British airways (2017)
  - Power surge failure
  - Back up system fails to engage
  - Information systems down at Gatwick and Heathrow
    - 1000 flights canceled over 3 days, 75000 people affected
    - 200M$ compensation costs
    - British Airways stock exchange loss 200M$

Critical failures

- ArianeV rocket launch failure, June 1996
- Due to a system / software defect

Why

Software as an industrial product

- Requires a large number of activities and tools, that must be coordinated and organized effectively

→ software process

Why

- All right, developing software is a f…g mess, and that is why i dont want to do it in my career

Why

SoftEng
http://softeng.polit

Why

- Software is everywhere!

Why

- Every person should be computer literate
  - Reading, writing, math, COMPUTING
  - Digital divide
    - Social and work exclusion of citizens who are computer illiterate
- Every engineer should be (way more than) computer literate

Why

- Every company should become a software company
  - Amazon vs. Border’s
  - Netflix vs. Blockbuster
  - Apple, Google vs. Toyota, Volkswagen, GM, ..

Why

- The economies of ALL developed nations are dependent on software.
- Expenditure on software represents a significant fraction of GNP in all developed countries.
- There is a clear relationship between innovation, productivity, wealth and expenditure in IT technology

Who

- Maurizio Morisio
  - Dept. Control and Computer Engineering
  - maurizio.morisio@polito.it
- Tommaso Fulcini
  - Tommaso.Fulcini@polito.it

When (and where)

Day	Room
Monday 8.30	R4
Wednesday 8.30	R4
Thursday 11.30	R2
Friday 11.30	R3

- BYOD (Bring Your Own Device)!
  - Classrooms have plugs
- All the lectures will be streamed and recorded

How

- Lectures and Exercises
  - In classroom, BYOD
- Participation is important
  - live / in streaming / on demand

How

GitLab

CATME
SMARTER Teamwork

zoom

What will I learn?

- Software Engineering
  - Motivation, issues, problems
- Requirement engineering
  - How to understand what the customer / user needs
- Design
  - How to put together components which deliver what the customer needs
- Verification and Validation
  - How to assure that the product does what it is supposed to do
- Project management
  - How to manage teamwork and deliver the product
- Software process
  - How to organize activities and tools, what process models are available

How to... pass the exam

- Two parts
  - Project work (explained in detail later)
  - Exam
- Final grade (max 33 points) =

  grade of project (max 15 points)

  +

  grade written exam (max 20 points).

  The project grade >= 12 /20

  In case the project is not done, max possible grade is 20/30
  30cum laude requires >= 32.5 points

Books and References

- Bruegge Dutoit, Object Oriented Software Engineering
- Ian Sommerville, Software Engineering, Pearson ed.
  - Also, Pfleeger, Pressman, Shach
- Martin Fowler, UML Distilled

And then?

- How does Software development work in practice?
  - Modern teams organize with agile practices
- Quality is a key component in engineering practice, how does it work in Software engineering?
  - Software analytics allow measuring and controlling Sw projects
- Evolution is intrinsic in software, how do you tame it?
  - Advanced debugging methods, log analysis, reverse engineering techniques

Software Engineering 2!

- Professors: Marco Torchiano, Antonio Vetrò
- A practical, project-oriented course to learn how software is crafted in modern teams
- Let’s try in practice
  - Software Scrum
  - Planning poker
  - Technical debt
  - Software smells

Project

Process and product

- Apply a state-of-the-art software engineering process to a (small) project:
  - Use of tools
  - Use of techniques
  - Use of process
  - Work in a team

- Work on a project, code, document, test and modify it

Constraints

- Project must be developed in parallel with course
- The object to be developed is the same for all the teams
  - Discussion inside the team is essential
  - Copying between teams is forbidden
- Antiplagiarism tools will be applied
- All communications only via defined tools
  - (mostly the Git repository, or Slack)

Steps

1. Teams will be defined by us
   - You cannot choose your teammates (as it happens in real life!)
2. Set up repository
   - You will receive an email (@polito account) for setting up your account and team repository
3. Access objects of development on repository
4. Produce various deliverables

To be produced

- Requirements document
- Design document
- Code
- Test suite – unit level, integration level
- Test suite, API level
- Change Request